

LAKSHIT RAJPUT

Senior SDET Engineer | AI-Driven Test Automation | Scalable QA Architecture & GenAI-Powered QA Engineering

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Designing resilient automation ecosystems that elevate software quality, engineering velocity, and release confidence at scale. Combining deep expertise in Playwright, API automation, cloud testing, and GenAI-powered QA workflows to build intelligent testing systems for modern digital platforms. Passionate about redefining quality engineering through AI-native automation, scalable framework architecture, and next-generation testing strategies that enable faster, smarter, and more reliable software delivery.

WORK EXPERIENCE,

SilverXis Inc

Senior Software Development Engineer in Test

Irving, TX

Dec 2022 – Present

- **Architected Enterprise QA Automation Platform:** Replaced a legacy Cypress framework with a scalable Playwright + TypeScript test ecosystem, enabling 3,000+ automated web/API test flows, reducing flaky failures from 22% to under 2%, and accelerating regression cycles by 65% through CI/CD-driven quality engineering.
- **Engineered Intelligent CI/CD Quality Gates:** Implemented GitHub Actions-based pipelines with Dockerized execution environments and distributed Playwright test sharding, reducing pipeline execution time from 3.5 hours to under 50 minutes while improving test stability and deployment confidence across Agile release cycles.
- **Built Robust API Automation Layer:** Designed REST Assured / Playwright-based API testing framework with SQL validation and contract testing, improving backend defect detection rate and enabling early shift-left validation across microservices-based architecture.
- **Designed Decentralized Quality Engineering Model:** Standardized SDET guardrails across 6 cross-functional Agile squads, accelerating feature delivery velocity by 20% and mitigating regression risks by 15% through upstream defect detection.
- **Scaled High-Velocity QA Automation Ecosystem:** Spearheaded the integration of AI failure analysis and test generation into CI/CD pipelines, optimizing 3,000+ test cases to reclaim 40% of manual triage effort and drive a 35% gain in signal accuracy.

Software Engineer in Test

ServiceLink (Fidelity National Financial)

Irving, TX

Jun 2022– Nov 2022

- **Rebuilt End-to-End Test Automation:** Replaced manual testing with an automated UI framework for a platform handling 500K+ monthly transactions, cutting manual QA effort by 80% and saving 320 engineering hours every month.
- **Built Automated API Test Suite:** Designed a modular testing system for 140+ REST endpoints, reaching 94% coverage and ensuring every code merge is automatically checked for bugs.
- **Streamlined Release and Compliance Workflows:** Implemented a BDD framework with clear documentation, allowing product and compliance teams to verify requirements instantly and cutting a 2-week release bottleneck.
- **Led Data Migration Testing Strategy:** Managed the quality and accuracy of migrating data from 3 legacy systems into a single database, writing over 200 validation scripts to prevent data loss and schema errors..

SilverXis Inc

Test Engineer

Irving, TX

June 2021 – Dec 2021

- Designed a Java/Selenium/TestNG/Maven test automation framework from 0→1, implementing Page Object Model and data-driven patterns; eliminated 2+ hours of daily manual regression and achieved 78% automation coverage within 6 months.
- Optimized Jenkins CI/CD pipeline configuration to enable parallel test execution across 4 agents; cut deployment validation time by 30%, accelerating release cadence from bi-weekly to weekly.
- Built iOS mobile test suite using Appium with dynamic element locators and XCUITest integration; increased mobile regression coverage by 60% and reduced escape defect rate for mobile releases.
- Conducted JMeter + LoadRunner performance benchmarks for API endpoints under simulated peak load (5,000 concurrent users); identified 2 critical bottlenecks resolved pre-launch, preventing estimated 40% revenue loss during peak traffic.

- Built and scaled a Cucumber/Gherkin BDD framework that automated 1,000+ regression test cases across a Barclays payment platform; saved 148 engineer-hours per monthly release cycle and shortened sprint testing lifecycle by 2 days — recognized with Accenture Excellence in Software Testing Award (Q1 2017).
- Optimized 350+ complex SQL queries for automated data validation against Mainframe DB2 and SQL Server, converting entirely manual data verification workflows into scheduled automated pipelines.
- Designed end-to-end test plans across 5 release phases for a high-volume payment processing system, achieving a 20% reduction in critical bugs escaping to UAT and reducing recurring production defects from an average of 5 to fewer than 2 per release.
- Leveraged HP-QC and performance analytics to optimize payment transaction processing time from 4 hours to under 100 seconds — a 99.6% reduction — through targeted load testing and bottleneck analysis.
- Led automation framework workshops for a 15-member engineering team; awarded 'Star of the Month' twice for outstanding delivery and cross-functional enablement impact.

AI & GENAI QUALITY ENGINEERING PROJECTS

AI Test Generation Engine — Built a LangChain + Claude API pipeline that ingests OpenAPI 3.0 specs and generates executable Playwright test skeletons with assertion logic, reducing test authoring time per endpoint from 45 minutes to under 8 minutes.

LLM Regression Testing Framework — Designed an evaluation harness using pytest and custom scoring functions to detect prompt regressions, output drift, and quality degradation across model versions, applied to internal GenAI feature releases.

Autonomous Flakiness Detector — Engineered a Python-based CI analytics tool that analyzes test run history, applies statistical anomaly detection, and auto-quarantines flaky tests with Slack alerting, reducing CI flakiness noise by 87%.

Quality Intelligence Dashboard — Built a Grafana dashboard consuming GitHub Actions test telemetry, surfacing release confidence scores, coverage deltas, and SLO burn rates for engineering leadership.

EDUCATION

The University of Texas at Dallas
MS, Information Technology & Management

Dallas, TX
Jun 2022

- Dean's Scholar (Top 20%); President, Data Science Club — led workshops and speaker series.
- Top Student Mentor for Data Science track — answered 800+ questions and coached 70+ students.

RN Shetty Institute of Technology
BS, Electronics & Instrumentation Engineering

Bangalore, IN
Jun 2015

SKILLS

Automation & Frameworks: Playwright, Cypress, Selenium, Appium, Pytest, TestNG, Pact, Karate

Languages & AI/GenAI: TypeScript, Python, Java, SQL, LLM Evaluation, AI-Assisted Test Generation, LangChain, Prompt Regression

CI/CD & Cloud: GitHub Actions, CircleCI, Jenkins, AWS (EC2, S3, Lambda), Docker, Kubernetes, Azure DevOps

Observability & Performance: Datadog, Grafana, k6, JMeter, Distributed Tracing, SLO Validation, RestAssured, Postman

Data & Methodologies: PostgreSQL, MongoDB, DynamoDB, ETL Validation, Shift-Left QE, BDD, Chaos Engineering, Agile/SAFe

CERTIFICATIONS & RECOGNITION

ISTQB Foundation Level Certified — International Software Testing Qualifications Board

Product Management Certificate — Product School

Accenture Excellence in Software Testing Award — Q1 2017 (Recognized for contributing to a 0-defect release)

Barclays/Accenture Star of the Month — July 2017 (×2 across tenure for automation impact)